

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

This assessment features a quick guide to the non-verbal reasoning tests, as well as an introduction to the rotation and reflection questions your child might encounter. Then, there are two practice tests for your child to have a go at. We have included an answer guide too so that you can help your child to identify any misconceptions.

What skills does this practise?

Non-Verbal Reasoning

Problem-Solving

Rotation

Reflection

Further Activity Ideas and Suggestions

For more non-verbal reasoning practice, check out this [11+ Non-Verbal Reasoning Analogies Assessment Pack](#) and this [11+ Non-Verbal Reasoning Week of Workouts](#). You can encourage your child to get in some bite-size practice of all the 11+ skills with this set of [11+ Activity Mats](#).

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Non-Verbal Reasoning



The non-verbal reasoning test involves solving problems that are represented with pictures and diagrams. There are a number of different types of questions for the non-verbal reasoning test which could include:

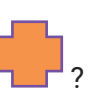
- finding shapes that are like each other based on similarities (for example colours and number of sides);
- finding the odd one out in a group of pictures;
- identifying a shape which has been rotated;
- breaking the code in a sequence of images;
- spotting the missing shape in a sequence;
- identifying the nets of 3D shapes.



Here is an example of the type of question you may get in this section:



1. Which is the correct code for figure 4?

 M B	 R B	 M Q	 ?
1	2	3	4

B	Q	B	R
Y	M	Q	Q
A	B	C	D

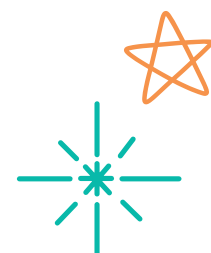


The answer would be D. The letter common to the first two shapes is B which represents the cross-hatching pattern they both have. Therefore Q represents the solid orange colour. The M is the letter representing the shape of the moon and it comes at the top of the box. Therefore the answer is D (R to represent the irregular dodecagon and the Q at the bottom to represent the solid orange colour).

The non-verbal reasoning test, for many children, can be very intimidating. The best way to prepare for this is through practice. Complete many different type of practice non-verbal reasoning tests with your child and discuss how each question is solved. This will give your child the experience needed to tackle the on-verbal reasoning tests.



For a thorough guide to all aspects of the 11+ test, try our **Survival Guide for Parents/Carers.**



11+ Non-Verbal Reasoning Practice Paper

Rotation and Reflection

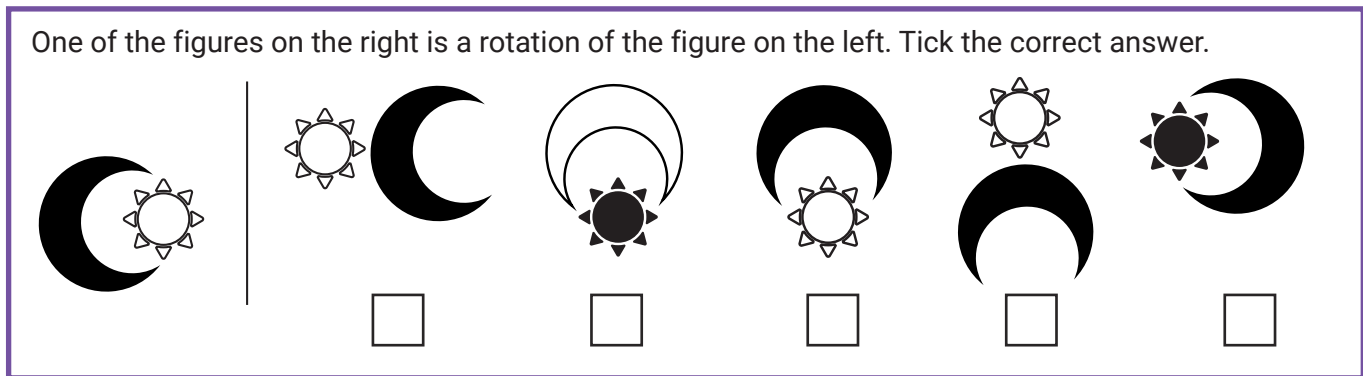
Many non-verbal reasoning papers include questions regarding rotation and reflection.

This question type requires you to identify increasingly complex figures after they have been rotated or reflected. You will need a keen eye and good observation skills as well as a strong understanding of the concept of rotation and reflection, which is usually taught in year 5 maths lessons.

Usually, you will be given a figure on the left followed by a selection of figures on the right. Your task is to choose the figure on the right which is a rotation or a reflection of the figure on the left. You might be asked to tick or circle the correct answer, or they may have been assigned letters such as a, b, c or d and you will be asked to write the corresponding letter.

Here is an example of a **rotation** question:

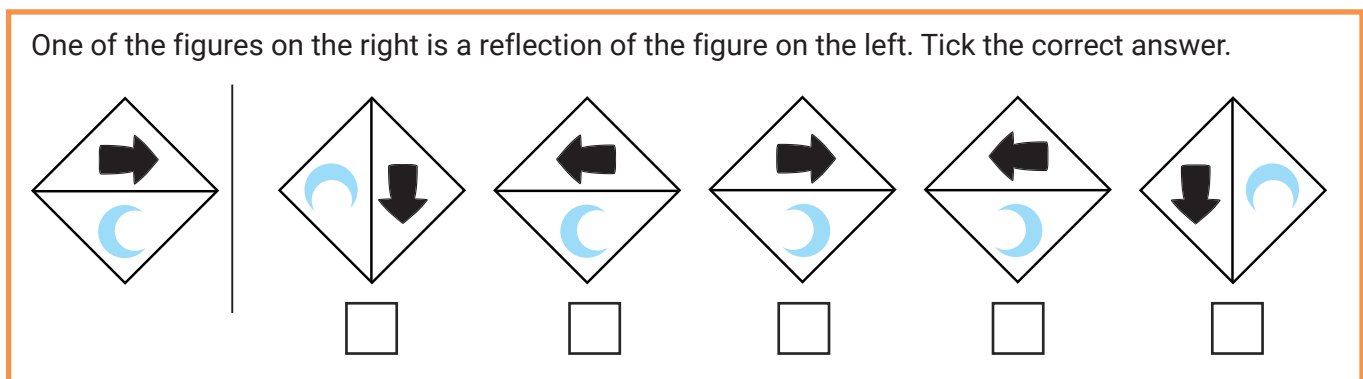
One of the figures on the right is a rotation of the figure on the left. Tick the correct answer.



The answer is option 3. It has been rotated 90° clockwise.

Here is an example of a reflection question.

One of the figures on the right is a reflection of the figure on the left. Tick the correct answer.



The answer is option 4. It is a reflection of the shape on the left.

In order to solve these problems, you will need to look at each element of the figure and carefully observe its features and orientation in order to determine if it is a rotation or a reflection of the original figure.

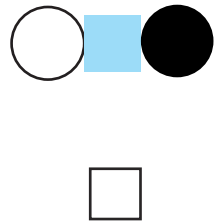
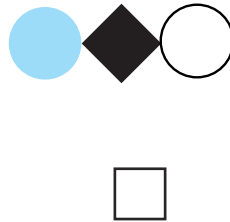
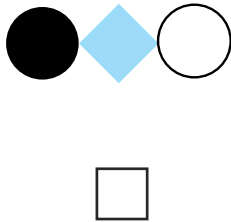
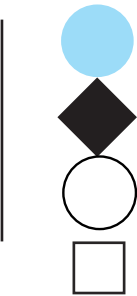
11+ Non-Verbal Reasoning Practice Paper

Rotation: Test 1

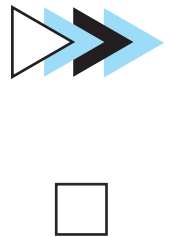
One of the figures on the right is a rotation of the figure on the left.

Tick the correct answer.

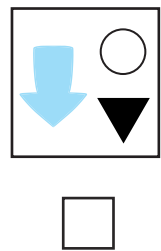
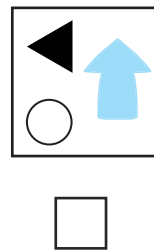
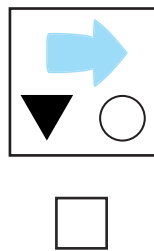
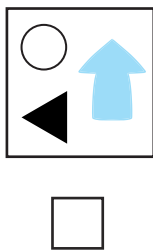
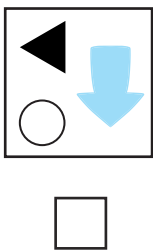
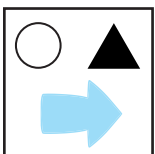
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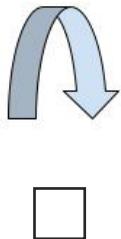
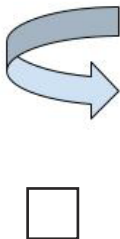
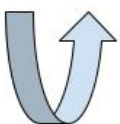
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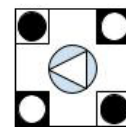
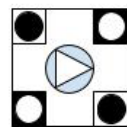
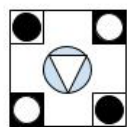
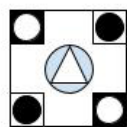
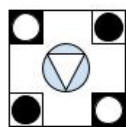
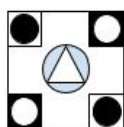
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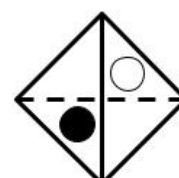
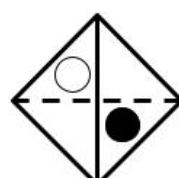
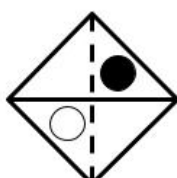
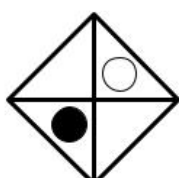
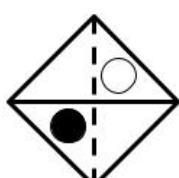
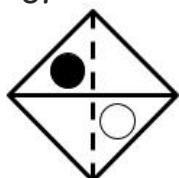
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11+ Non-Verbal Reasoning Practice Paper:

Rotation: Test 1 Answers

1.

Options:

- ☐ Option 1: Black circle, light blue diamond, white circle (original)
- ☒ Option 2: Light blue circle, black diamond, white circle
- ☐ Option 3: Black circle, light blue diamond, white circle (original)
- ☐ Option 4: Light blue circle, black diamond, white circle
- ☐ Option 5: White circle, light blue diamond, black circle
- ☐ Option 6: White circle, light blue diamond, black circle

The correct answer is option 2. It has been rotated 90° anticlockwise.

2.

Options:

- ☒ Option 1: Black triangle, white triangle, light blue triangle
- ☐ Option 2: Black triangle, white triangle, light blue triangle
- ☐ Option 3: Black triangle, white triangle, light blue triangle
- ☐ Option 4: Black triangle, white triangle, light blue triangle
- ☐ Option 5: Black triangle, white triangle, light blue triangle
- ☐ Option 6: Black triangle, white triangle, light blue triangle

The correct answer is option 1. It has been rotated 180°.

3.

Options:

- ☐ Option 1: White circle, black triangle, light blue arrow pointing right
- ☐ Option 2: White circle, black triangle, light blue arrow pointing right
- ☐ Option 3: White circle, black triangle, light blue arrow pointing right
- ☒ Option 4: White circle, black triangle, light blue arrow pointing up
- ☐ Option 5: White circle, black triangle, light blue arrow pointing right
- ☐ Option 6: White circle, black triangle, light blue arrow pointing right

The correct answer is option 4. It has been rotated 90° anticlockwise.

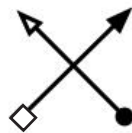
4.

Options:

- ☒ Option 1: Light blue arrow pointing up and to the right
- ☐ Option 2: Light blue arrow pointing up and to the right
- ☐ Option 3: Light blue arrow pointing up and to the right
- ☐ Option 4: Light blue arrow pointing up and to the right
- ☐ Option 5: Light blue arrow pointing up and to the right
- ☐ Option 6: Light blue arrow pointing up and to the right

The correct answer is option 1. It has been rotated 90° clockwise.

5.



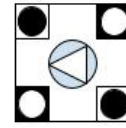
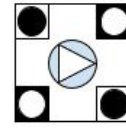
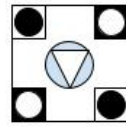
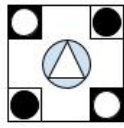
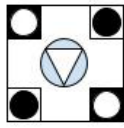
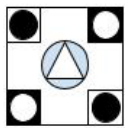
The correct answer is option 5. It has been rotated 180°.

6.



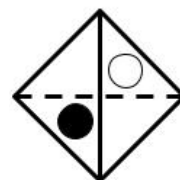
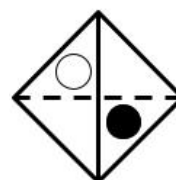
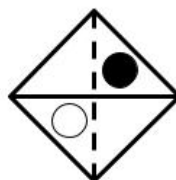
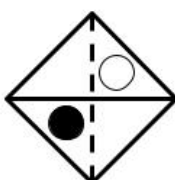
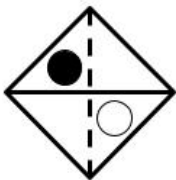
The correct answer is option 3. It has been rotated 90° clockwise.

7.



The correct answer is option 3. It has been rotated 180°.

8.



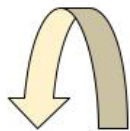
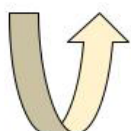
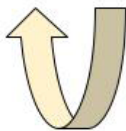
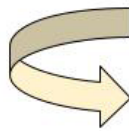
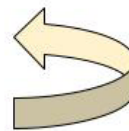
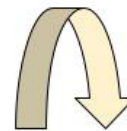
The correct answer is option 5. It has been rotated 90° anti-clockwise.

11+ Non-Verbal Reasoning Practice Paper

Reflection: Test 2

The figure on the left has been reflected in the black line. Tick the correct reflected figure.

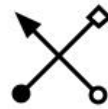
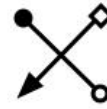
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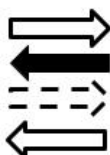
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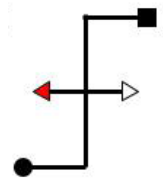
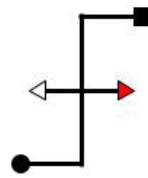
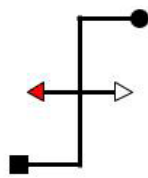
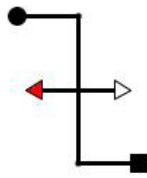
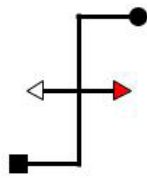
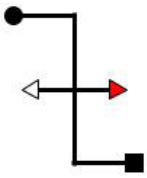
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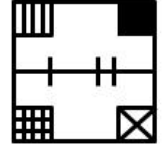
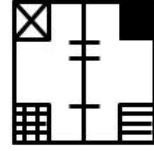
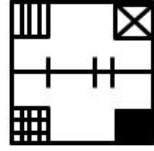
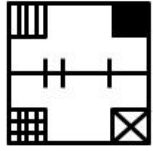
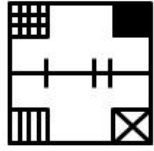
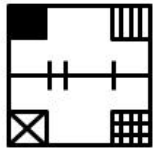
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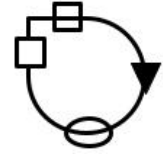
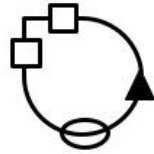
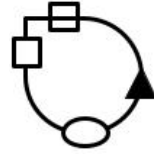
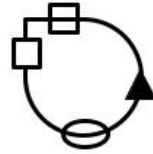
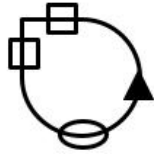
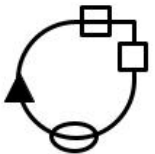
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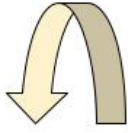
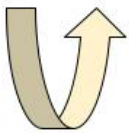
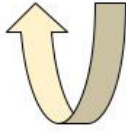
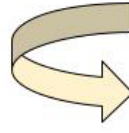
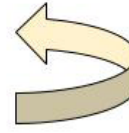
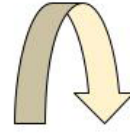
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11+ Non-Verbal Reasoning Practice Paper:

Reflection: Test 2 Answers

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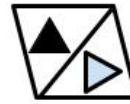
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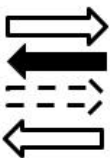

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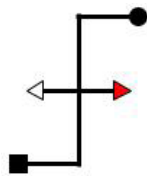
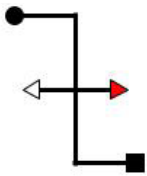
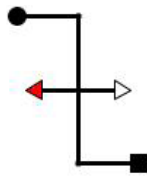
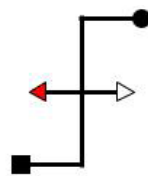
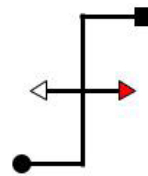
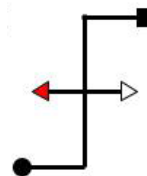
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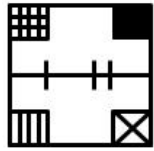
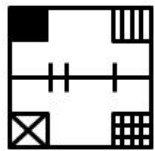
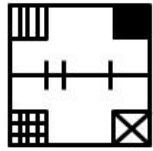
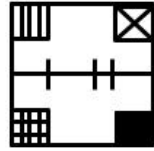
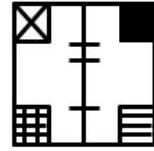
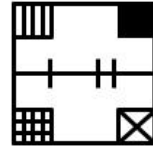
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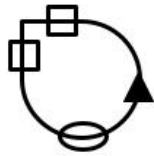
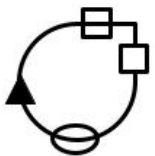
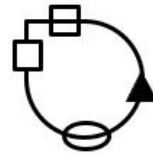
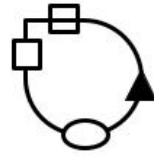
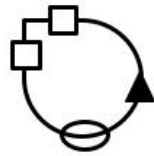
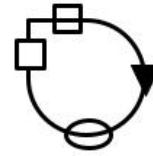
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